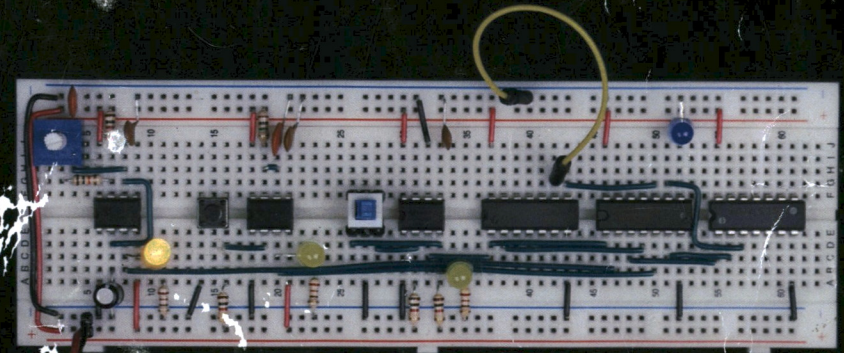
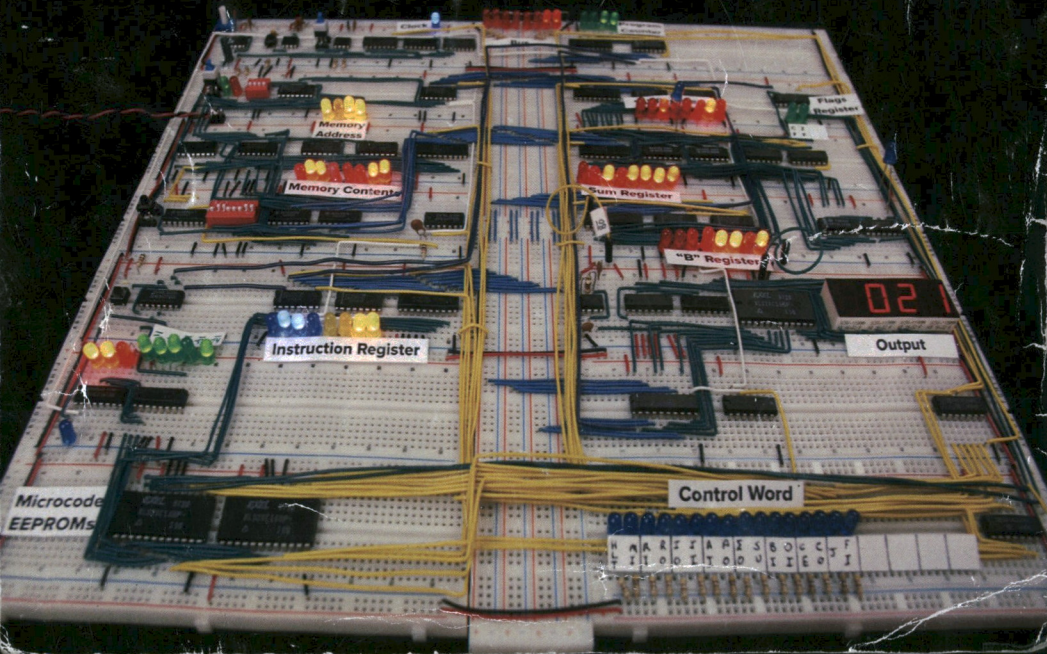


8-bit breadboard computer

Kit 1 of 4

Clock module

www.eater.net
youtube.com/beneater



Ben Eater

Welcome!

This is the first kit in a set of four that, along with my series of videos on YouTube, will help you develop a deep understanding of how programmable CPUs really work. I hope when you're done, you'll know enough to extend the computer design and make it your own.

With this first kit, you'll build the clock module. The computer's clock is used to synchronize all operations. The clock we're building in this kit is based on the popular 555 timer IC, and as part of this project you'll use the 555 in three different ways: astable, monostable, and bistable.

The computer clock is adjustable-speed (from less than 1Hz to a few hundred Hz). It can also be put into a manual mode where you push a button to advance each clock cycle. This will be a really useful feature for debugging the computer later on.

Where are the instructions to build this kit?

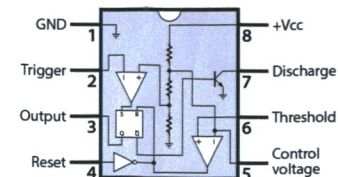
Surprise! This booklet doesn't actually contain the full instructions for building the kit. This kit is designed as a companion to my series of YouTube videos on building a computer from scratch. You should watch the videos and follow along yourself. I hope that by *not* having a detailed "paint-by-numbers" instruction book, you'll learn more about how the circuits really work. But rest assured, the videos are quite detailed and should contain everything you need to be successful:

The video instructions for this kit are at
www.eater.net/8bit/clock

Integrated circuit quick reference

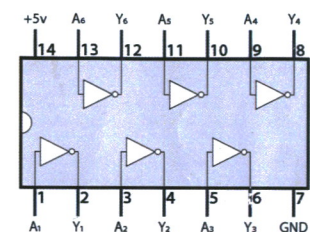
555 — Timer

The 555 can act as either a timer, an oscillator, or a latch. In this project, we use all three modes to build our computer's clock.



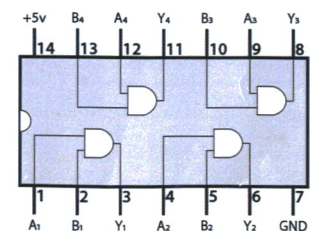
74LS04 — Inverter

A	Y
0	1
1	0



74LS08 — AND gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



74LS32 — OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

